

WASP-43b

R-2560

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140316 · created 2026-08-26T23:04:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456732.81144 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.018
Transit depth (Rp/Rs)^2	3.91 +/- 0.72 [%]
Orbital Inclination (inc)	80.78 +/- 1.34
Airmass coefficient 1 (a1)	1.003 +/- 0.027
Airmass coefficient 2 (a2)	0.0 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.65 %
Best Comparison Star	#1 - [199, 65]
Optimal Method	PSF photometry
Transit Duration (day)	0.0487 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.198 ± 0.018 vs 0.1594 (deviation 1.83σ)
Duration fit vs published	0.0487 d vs 0.0512 d (deviation 0.38σ)
Mid-time O–C	1.6 min
Depth SNR	9.4
Residual scatter	2.65 %

Light curve

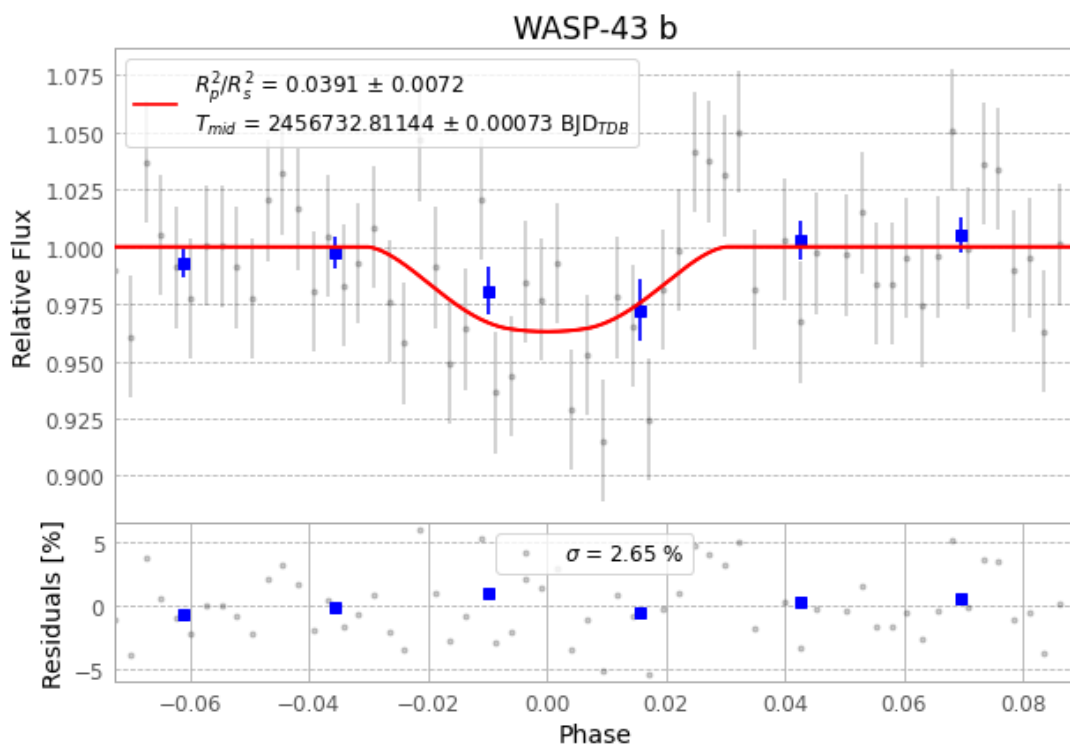


Figure 1 — FinalLightCurve_WASP-43 b_2014-03-15.png (SHA-256 d4f943fe54b680a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9c3008abf7304fde...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140316055412.FITS → WASP-43140316090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140316.log (204b1ee25916...), FinalLightCurve_WASP-43 b_2014-03-15.png (d4f943fe54b6...), FinalLightCurve_WASP-43 b_2014-03-15.csv (d8ceafe74795...)

Compute resources

Wall time 117.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:04Z.